

□ protected mode

Protected mode-র physical address এর
সীত address বের করার।

And fir introduce ~~पर~~ ~~इसे~~ physical व
सिद्ध problem हिला पर

Problem:

- i) Physical सि $2^{20} \rightarrow 1\text{MB}$, सार program 1MB सि सकि रह वाव ना।
- ii) At a time सिरे सकि program run सकि ना।
- iii) Offset (0000 - FFFF) सि सकि रह ना।
 $\rightarrow 16 \text{ bit}$
 $2^{16} = 2^{10} \times 2^6$
 $= 64 \text{ KB}$

(ii) ~~It is not~~ the transmission segment

→ for problem solve in protected mode & memory management unit we use MMU.

fig. 10.24 portion access with write
set 25.

→ mmu to virtual memory Address concept use

swap in, swap out

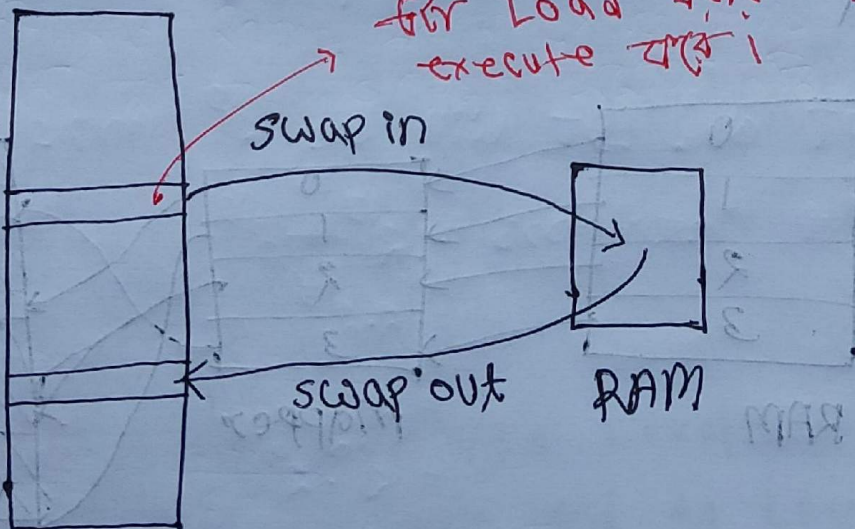
swap in:

secondary memory me RAM ke program ke specific portion Load kr.

swap out:

program me RAM ke specific portion Load kr kr secondary memory me save/write kr kr. kr kr edited Data save kr kr.

Ex:

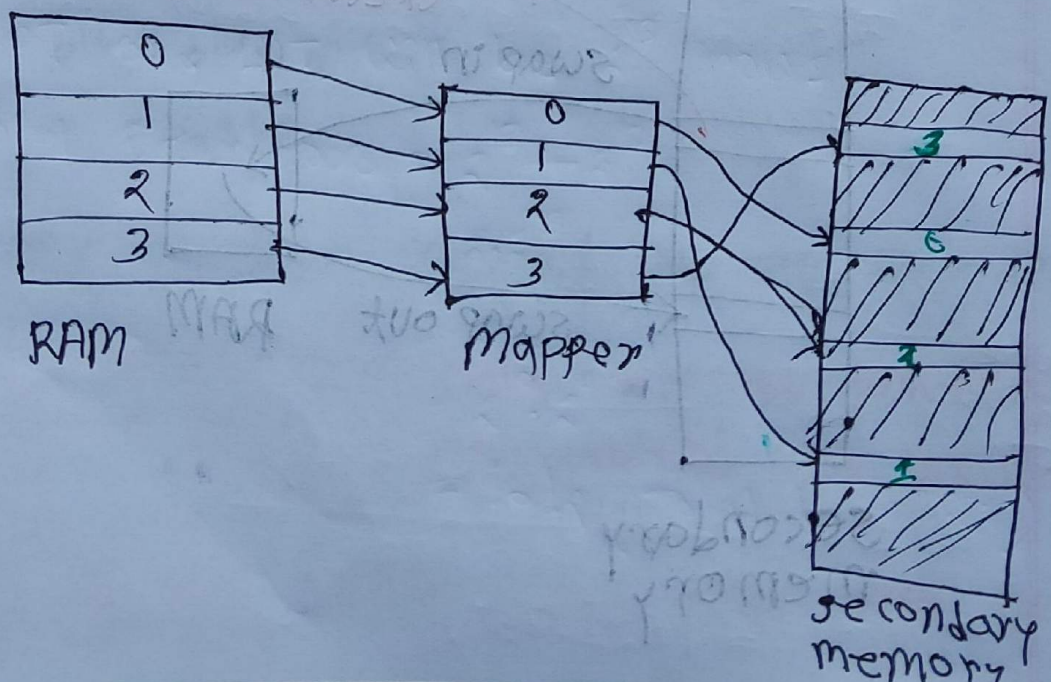


Advantage of virtual memory:

- i) solve limited physical memory size
- ii) solve discontinuous memory address problem
- iii) provide program protection

Translator/mapper

Translator mainly has table which maps secondary memory to main program address and (see secondary to program sequentially or vice versa) RAM translator for mapping secondary memory to program load.



- Mapper for access protection (user mode, or kernel mode level) for each program access or user access
- (Mapper > 1mb)
- Mapper for virtual memory.

Real mode:

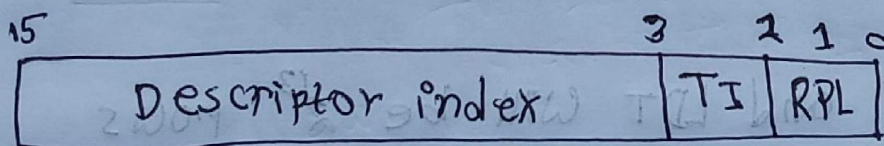
Physical = segment * 10H + offset

Segment Register contains segment Address

Protected mode

Segment Register contains selector.

Selector:



~~1) Descriptor index~~

2) RPL

- Request privilege Level
- 00 is highest and 11 Lowest

mainly for access and mapper for or indicate that

00 $\longrightarrow$ highest (kernel level)
 01 $\longrightarrow$ service level
 10 $\longrightarrow$ service level
 11 $\longrightarrow$ lowest (Application Level)

ii) TI

$\longrightarrow$ Table indicator

$\longrightarrow$ TI = 0, global description table (GDT)
 $\longrightarrow$ TI = 1, Local descriptor table (LDT)

iii) Descriptor index

$\longrightarrow$ 13 bit index
 $\longrightarrow 2^{13} = 8192$ descriptors

GDT and LDT will be 2^{13} rows

$$\begin{aligned}
 \text{Total Descriptors} &= 2^{13} + 2^{13} \\
 &= 2 \cdot 2^{13} \\
 &= 2^{14} \\
 &= 2^{10} \cdot 2^4 \\
 &= 16K \text{ (Total 16 MB)}
 \end{aligned}$$

Exercise

#29: if the DS register contains 0020 H in a protected mode system, which global descriptor table entry is accessed.

DS = 0020 H

= 0000 0006 0010 0000

Descriptor index

always 9 bit numbering

RPL

TI = 0, GDT

= 2 0009 (th entry)

#30: if DS = 0103 H in protected mode system, the requested privilege level?

DS = 0103 H

= 0000 0001 0000 0011

- i) Application level (Lowest)
- ii) Global
- iii) ~~0020H~~ 0020 (20th entry)

#31:

if DS = 0105 H in a protected mode system, which entry, table, and requested privilege level are selected?

DS = 0105 H

= 0000 0001 0000 0101

→ serv

i) 0020 th entry

ii) TI = 1 (LDT)

iii) RPL = 01 (service level)

23th class

GDT, LDT manage કરા OS નિષ્ણ ।

Descriptor table:

Descriptor table ઠી ડાટા segment ની information આપે ।

Access	Limit
Base Address	

i) Access

તે જો access કરી શકે છે તે સલા આપે

ii) Base Address

જો segment ક્યાં રહે છે તે બતાવે ।

iii) Limit

to length of a segment.

80286

24 bit base Address and Limit = 16 bit

→ segment can start from 0 to $(2^{24}-1) = 16 \text{ MB}$

- segment size can be from 1 to $2^{16} = 64k$

→ চিহ্নের সংখ্যার 0 to $(64k-1)$ ২য় স, যার
বিপরীত বিপরীত সংখ্যক।

80386

32 bit base Address and Limit = 20 bit

→ segment can be start from 0 to $\binom{32-1}{2} = 9 \text{ GB}$

→ Segment size can be from 1 to $2^{20} = 1\text{MB}$

Granularity bit! (G)

$G=0$ হলে limit প্রাপ্তি byte-এ শিখাব যথাক্রমে:

→ segment can be start from 0 to $(2^{32}-1) = 4 \text{ GB}$

→ segment size can be from 1 to $2^{20} = 1\text{MB}$

$G=1$ Each cell contains 1 kb

→ segment size can be ~~1x1KB~~ 1MB x 4KB

$$= 2^{20} \times 4 \times 2^{10}$$

$$= 2^{20} \times 2^2 \times 2^{10}$$

$$= 4 \text{ GB}$$

N.B

Descriptor table is kind of table slip or slip which shows the location of program. This shows slip shows the location of course for which program is running.

- Global Descriptor table (GDT) is applicable; it has one row which is called as global/system descriptor.
- Local Descriptor table (LDT) is data specific program for which; it has one row which is called as Local/Application register.
- Each table descriptor is 8 byte long; table has maximum length 64KB.

Each table has 2^{13} indexes & each index is 8 bytes long.

$$\text{So length} = (2^{13} \times 8) \text{ B}$$

$$= 2^{16} \text{ B}$$

$$= 2^{10} \cdot 2^6 \text{ B}$$

$$= 64 \text{ KB}$$

→ हर 8 byte or 64 bit विशेष आकार का निरूपित किया जाता है:

0000 0000 0000 0000	Access right 8 bit	Base B23 (Last 8 bit) B16
Base (First 16 bit)	Limit 16 bit	
B15	B0 L15	L0

80286

Base (8 bit) B31 B24	G	D	O	A	Limit (4 bit) L19 L16	Access rights (8 bit)	Base (8 bit) B23 B16
B15	Base (16 bit)				B0 L15	Limit	L0

80386

⇒ G = 0

Base = start

Physical Address = Base + offset

End = Base + limit

⇒ G = 1

Base = start

step size.

End = Base + limit * 4k + 4k - 1

4k is changeable and it is in decimal.

Ex: slide 55

i) segment register = AC6 H

Ad 2 = 1010 1100 0110
Des index TI RPL

$$RPL = (10)_2 = (2)_{10}$$

$$TI = 1 \rightarrow LDT$$

$$DI = \underline{0001} \underline{0101} \underline{1000}$$

1 5 8

$$= 158 H$$

SO, 158 H entry of LDT is selected.

ii) Base = C002A00E H

Limit = 2A043 H

Access = B4H, G = 1

starting address = ~~E0~~ Base

$$= C002A00E H$$

Ending Address = Base + limit * 4k + 4k - 1

$$= C002A00E + 2A043 * (1000)$$

$$+ 1000 - 1$$

$$= EAO6E00D H$$

iii) physical Address = Base + offset
= C002A00E + 0051B
= C00EA519 H

Access Rights Byte:

- Controls the access to the protected mode segment.
- If the segment grows beyond its limit, the OS is interrupted indicating a General protection fault.
- We can specify ~~the~~ whether a data segment can be written or is written-protected.

P	DPL	S	E	ED/C	R/W	A
7	6 5	4	3	2	1	0

P → ~~segment~~ RAM to Load ~~segment~~ ~~segment~~ segment
or ~~segment~~ ~~segment~~ present.

$p = 0$, segment not present

$p = 1$, segment exist & it has a valid Base & Limit

DPL → Descriptor Privilege level (00 → highest
11 → lowest)

A → Accessed bit

0 → Not accessed yet

1 → Already accessed or Access granted.

R/W $\rightarrow$ Read and write

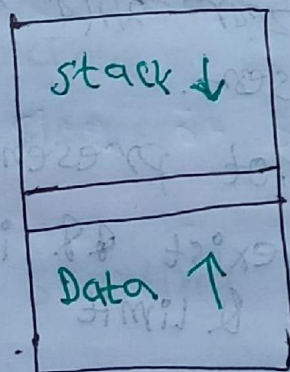
Data ($E=0$)	code ($E=1$)
$W=0$; Read only	$R=0$ (Only execute)
$W=1$; Read + Write	$R=1$ (Execute & read)

E $\rightarrow$ Executable or not.

$E=1$; code segment

$E=0$; Data/stack segment.

ED/C bit $\rightarrow$ Expand down



1 $\rightarrow$ stack (આવડે સ્ટેક તરફ આસપાસ Data આમલ)

0 $\rightarrow$ Data (Data ને તરફ segment તરફ આમલ આસપાસ)

S $\rightarrow$ Descriptor type bit

$S=0 \rightarrow$ System/global descriptor table

$S=1 \rightarrow$ code/Data/Local " "

Ex: slide: 55

iv) ~~B4~~ Access = ~~B4~~ H

= 1011 0100

P → Access bit
E → ED/c

So, ED/c = 1 (stack segment)

v) A = 0, so Access not granted

vii) P = 1, segment is exist in the RAM so defined

24th class

i) Segment size = $(\text{limit} + 1)_{10}$ byte; $g = 0$
Segment size = $(\text{limit} + 1)_{10} \times 4k$ byte; $g = 1$

ii) Total # of descriptors of a program
= total descriptor in LDT + total descriptor in GDT

$$= 2^{13} + 2^{13}$$

$$= 2 \cdot 2^{13}$$

$$= 2^{14}$$

→ Also total # of segment possible.

iii) Total virtual size of that program
= # of descriptors $\times$ size of segment

$$\begin{aligned} &= 2^{14} \times (\text{limit} + 1)_{10} \text{ byte ; } g=0 \\ &= 2^{14} \times (\text{limit} + 1)_{10} \times 4k \text{ byte ; } g=1 \end{aligned}$$

Ex:

if descriptor index = 10 bit, Limit = 1FF
step size = 8k, virtual memory size?

$$\text{Total descriptors} = 2^{10} + 2^{10}$$

$$= 2^{11}$$

$$\text{size of segment} = (1FF + 1)_{16} \times 8k \text{ Byte}$$

$$= (200)_{16} \times 8k \text{ Byte}$$

$$= (512)_{10} \times 8k \text{ Byte}$$

$$= 2^9 \times 2^3 \times 2^{10} \text{ Byte}$$

$$= 2^2 \times 2^{20} \text{ Byte}$$

$$= 4MB$$

જો આ મંદિ ટિપ્પર ડેસ્ક્રિપ્ટર ટિપ્પર લિમિટ બદલાય
અને આ ટિપ્પર લિમિટ ફિક્સ થાય

if 80386 $\rightarrow$ limit = FFFF

if 80386 $\rightarrow$ limit = FFFFFFFF

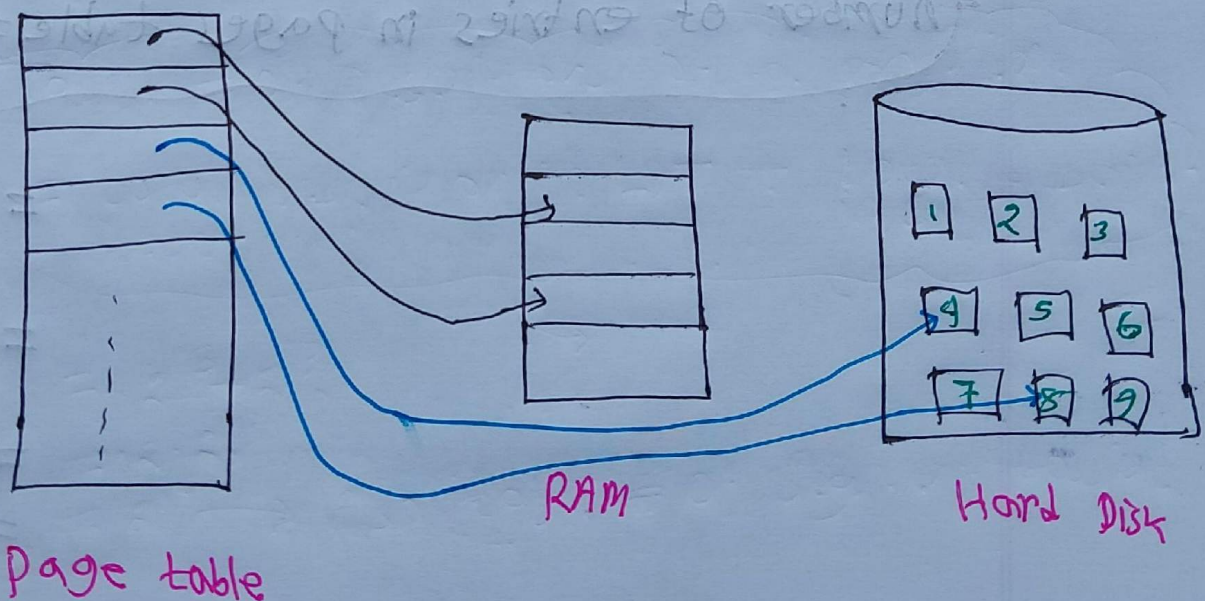
size of virtual memory: Total descriptor $\times$ size of segment

$$\begin{aligned} &= 2^{11} \times 4MB \\ &= 2^{11} \times 2^2 \times 2^{30} B \\ &= 2^3 \cdot 2^{30} B \\ &= 8 GB \end{aligned}$$

paging

For segment for variable we use it. But now size is fixed of a page.

Page table $\rightarrow$ mapper.



~~Page size~~

$$\text{Page table size} = \frac{\# \text{ of pages in page table}}{\text{virtual address}} \times \text{size of a page entry}$$

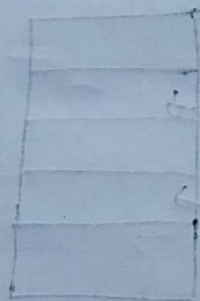
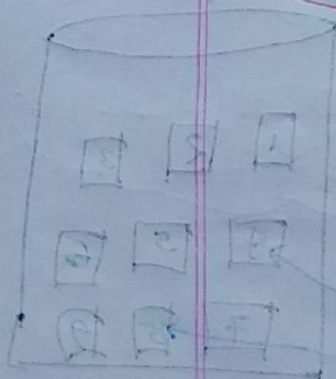
How many page store = 9

Summer Q3 (C)

Page table size = 1KB

entry size = 2B

$$\text{Number of entries in pages table} = \frac{\text{table size}}{\text{entry size}}$$



$$\begin{aligned} &= \frac{1\text{KB}}{2\text{B}} \\ &= \frac{1024\text{B}}{2\text{B}} \\ &= 512 \end{aligned}$$

= 512 pages in RAM.

so, number of page in secondary memory
 $= \text{Total virtual page} - \# \text{ of page in RAM}$

$$= 2048 - 512$$

$$= 1536$$

if two different question

Q How much space has RAM allocated for the program? $\forall$ Total virtual memory size? 512 pages in RAM, 2048 Total virtual pages, 1 page = 8MB

1 page = 8MB

$$\begin{aligned} 512 \text{ " } &= 512 \times 8 \text{ MB} \\ &= 512 \times 8 \times 2^{20} \text{ B} \\ &= 2^9 \times 2^3 \times 2^{20} \text{ B} \\ &= 4 \text{ GB} \end{aligned}$$

1 page $\rightarrow$ 8MB

$$\begin{aligned} 2048 &\rightarrow 2048 \times 8 \text{ MB} \\ &= 2^{11} \times 2^3 \times 2^{20} \text{ B} \\ &= 16 \text{ GB} \end{aligned}$$